

Changes in laboratory variables in rheumatoid arthritis patients during a trial of fasting and one-year vegetarian diet.

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We have previously reported that significant improvement may be obtained in rheumatoid arthritis patients by fasting followed by a vegetarian diet for one year. The present study was carried out to examine to what extent biochemical and immunological variables changed during the clinical trial of fasting and vegetarian diet. For the patients who were randomised to the vegetarian diet there was a significant decrease in platelet count, leukocyte count, calprotectin, total IgG, IgM rheumatoid factor (RF), C3-activation products, and the complement components C3 and C4 after one month of treatment. None of the measured parameters changed significantly during this period in the group of omnivores. The course of 14 of 15 measured variables favored the vegetarians compared with the omnivores, but the difference was only significant for leukocyte count, IgM RF, and the complement components C3 and C4. Most of the laboratory variables declined considerably in the vegetarians who improved according to clinical variables, indicating a substantial reduction in inflammatory activity. The leukocyte count, however, decreased in the vegetarians irrespective of the clinical results. Thus, the decline in leukocyte count may be attributed to vegetarian diet per se and not to the reduction in disease activity. The results of the present study are in accordance with the findings from the clinical trial, namely that dietary treatment can reduce the disease activity in some patients with rheumatoid arthritis.